

Your grade: 96%

Your latest: 96% • Your highest: 96% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding dimensionality reduction, Vision Transformers, and PyTorch Lightning modules. However, there is room for growth in understanding Graph Neural Networks. Before retrying this assignment, we recommend focusing on this key area:

- **Graph Neural Networks (101):** In Question 2, your understanding of the advantages of using Graph Neural Networks (GNNs) was incomplete. Try reviewing [this item](#) to refresh your memory.

Good job on the rest of the assignment!

Retry assignment



1. What is the purpose of dimensionality reduction techniques in visualizing model results?

1 / 1 point

- ☐ To enhance the accuracy of the model.
- ☒ To make the data easier to interpret visually by reducing it to 2D or 3D.
- ☐ To remove noise from the dataset.
- ☐ To reduce the size of the dataset for faster computation.

✔ Correct

Correct! Dimensionality reduction helps in visualizing high-dimensional data in 2D or 3D.

2. Which of the following are advantages of using Graph Neural Networks (GNNs)?

0.8 / 1 point

- ☐ Simplified computation compared to traditional methods.
- ☐ Inability to scale to large graphs.
- ☒ Superior performance on node classification tasks.

✔ Correct

Correct! GNNs have shown superior performance on various node classification tasks.

- ☐ Ability to handle dynamic graphs.
- ☒ Enhanced feature extraction from node attributes.

✔ Correct

Correct! GNNs can extract features from node attributes effectively.

You didn't select all the correct answers

3. Which of the following are components of a Vision Transformer (ViT)?

1 / 1 point

- ☒ Image patches

✔ Correct

Correct! Vision Transformers split images into patches as part of their processing pipeline.

- ☐ Convolutional layers
- ☐ Recurrent layers
- ☒ Position embeddings

✔ Correct

Correct! Position embeddings are used in Vision Transformers to retain the spatial information of image patches.

- ☒ Multihead self-attention

✔ Correct

Correct! Multihead self-attention is a key component in Vision Transformers, allowing the model to focus on different parts of an image simultaneously.

4. What are some benefits of using Vision Transformers in computer vision tasks?

1 / 1 point

- ☐ They are less computationally expensive than CNNs.
- ☐ They require very little data for training.
- ☒ They can handle long-range dependencies in images.

✔ Correct

Correct! Vision Transformers can handle long-range dependencies in images due to their self-attention mechanism.

- ☐ They are inherently better at image classification than any other model.
- ☒ They can leverage large-scale pre-trained models.

✔ Correct

Correct! Vision Transformers can benefit from large-scale pre-trained models, which can improve their performance on various tasks.

5. Which function must be defined in a PyTorch Lightning module to perform a forward pass?

1 / 1 point

- ☐ configure_optimizers
- ☐ validation_epoch_end
- ☐ train_step
- ☒ forward

✔ Correct

Correct! The forward function is essential for defining the computation performed at every call.